

COVID-19 Global Impact Dashboard — Executive Summary

Objective: Analyze cases, deaths and vaccination rollout over time, normalize comparisons by population, and communicate trends through an interactive dashboard.

Key findings

Area	Finding / action
Trend	Use 7-day rolling averages to reveal wave intensity while reducing reporting noise.
Country comparison	Use cases/deaths per million rather than raw totals to make comparisons fairer.
Vaccination	Compare vaccination coverage with mortality descriptively; do not claim causality.
Risk metric	Case fatality rate and doubling time add context beyond simple counts.
Dashboard	KPI cards + trend line + country bars + vaccination scatter + slicers.

1. Major COVID waves are easier to compare after smoothing daily reporting noise.
2. Population-normalized metrics change country rankings compared with raw totals.
3. Vaccination rollout provides an important timeline for interpreting later mortality patterns.
4. Outliers should be investigated using demographic, testing, reporting and healthcare context.
5. The dashboard is designed for a non-technical audience: headline KPIs first, then trends and country comparisons.

Production refresh: Run `src/download_owid_data.py` and `src/clean_owid_data.py` before final submission so the dashboard uses the real OWID public dataset.

Source: Our World in Data COVID-19 Data Explorer — <https://ourworldindata.org/explorers/covid>